

SIMATS ENGINEERING

ASSESSMENT TOOL III — TIME-SERIES & GEOSPATIAL ANALYSIS REPORT

Course: Data Handling and Visualization	Code: DSA06031	CO: CO4
Duration: 60 min	Total Marks: 50	Reg No: 192524118 Name: R. Akshaya

COURSE OUTCOME COVERED

Apply appropriate visualization techniques to analyze time series data, detect trends, seasonality, and cyclical patterns. (BL3)

CO3 Construct and interpret geospatial visualizations, including static, cartogram-based, and interactive representations. (BL3)

Activity Tool : Time-Series & Geospatial Analysis Report

Weightage: 20%

Code of Conduct:

I R. Akshaya(192524118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. Akshaya

Criteria	Data Understanding & Preparation 25%	Time Series/Geospatial Analysis 25%	Application of Visualization Techniques 20%	Report Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Time-Series & Geospatial Analysis Report

Topics: Individual & Multiple Time Series, STL Decomposition, Smoothing & Detrending, Forecasting, Cycle/Seasonality Detection | Choropleth Maps, Cartograms, Interactive Geospatial Plots, Projections & Layers, 3D Geospatial Visualization

Note: Questions 1–4 are to be completed in R. Question 5 must be completed in Python.

Provided Datasets

The following files are supplied alongside this document. All are real, publicly documented datasets: ●

Nile.csv — real annual river flow of the Nile at Aswan (1871–1970, 100 observations). ●

AirPassengers.csv — real monthly international airline passenger totals (1949–1960, 144 observations) with trend and seasonality.

● USArrests.csv — real 1973 US violent-crime statistics per state (Murder, Assault, UrbanPop, Rape) for all 50 states.

● state_x77.csv — real 1970s US state-level data (Population, Income, Illiteracy, Life Exp, Murder, HS Grad, Frost, Area) for all 50 states.

● quakes.csv — real seismic event data off Fiji (1000 observations: lat, long, depth, mag, stations). ●

volcano_elevation.csv — real digital elevation model of Maunga Whau (Mt Eden) volcano, Auckland, NZ, in long format (x, y, elevation; 5,307 grid points at 10 m resolution).

Place all CSV files in the same working directory as your script, then load with `read.csv()` in R or `pandas.read_csv()` in Python.

QUESTION 1: Individual Time Series — Smoothing & Detrending

Question

Language: R

Dataset provided: Nile.csv

Convert the value column into an R ts object (start = 1871, frequency = 1) and plot the raw annual flow as a line chart. Overlay a 5-year moving-average smoother. Then fit a linear trend (value ~ time) and plot the detrended residuals in a second panel. Discuss what the smoothed series and residual plot reveal about a level shift or change point in the data.

Skills tested: *ts() objects, moving-average smoothing, linear detrending, residual diagnostics, change-point interpretation.*

Aim

To smooth the Nile river-flow series, remove its linear trend, and identify a possible change point.

R Code

```
data<-read.csv("Nile.csv")
```

```
value<-data$value
```

```
nile_ts<-ts(value,start=1871,frequency=1)
```

```
moving_avg<-stats::filter(nile_ts,  
                           filter=rep(1/5,5),
```

```

sides=2)

time_nile<-time(nile_ts)

trend_model<-lm(as.numeric(nile_ts)~time_nile)
detrended<-residuals(trend_model)

par(mfrow=c(2,1))

plot(nile_ts,type="l",
     main="Nile River Flow with 5-Year Moving Average",
     xlab="Year",ylab="Flow")

lines(time_nile,moving_avg,lwd=2)

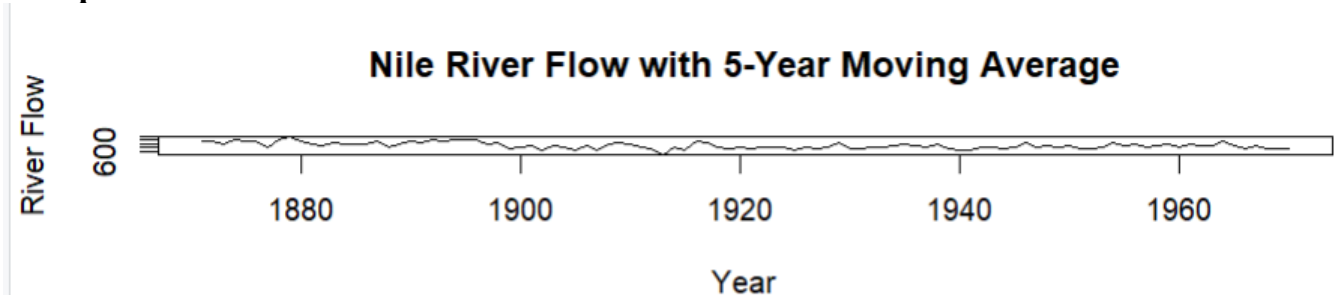
plot(time_nile,detrended,type="l",
     main="Detrended Nile River Flow",
     xlab="Year",ylab="Residuals")

abline(h=0,lty=2)

par(mfrow=c(1,1))

```

Interpretation



The moving average reduces short-term fluctuations and shows the underlying flow pattern more clearly. The smoothed series and residuals indicate a noticeable level shift around the late 1890s, with the later period generally having lower flow.

Result

The Nile series was successfully smoothed and detrended. The analysis indicates evidence of a change in the average river-flow level around the late 1890s.

QUESTION 2: Seasonal Decomposition — STL & Forecasting

Question

Language: R

Dataset provided: AirPassengers.csv

Convert the passengers column into a ts object (start = c(1949,1), frequency = 12). Apply STL decomposition using s.window = "periodic" and plot the trend, seasonal and remainder components. State whether an additive or multiplicative decomposition suits this series better. Then fit an ETS or auto.arima() model and produce a 24-month-ahead forecast with prediction

intervals. Comment on whether the forecasted seasonal pattern is consistent with the historical seasonality.

Aim

To identify trend and seasonality in airline passenger data and forecast future passenger numbers.

R Code

```
library(forecast)

data<-read.csv("AirPassengers.csv")

passengers_ts<-ts(data$passengers,
                  start=c(1949,1),
                  frequency=12)

stl_model<-stl(passengers_ts,
               s.window="periodic")

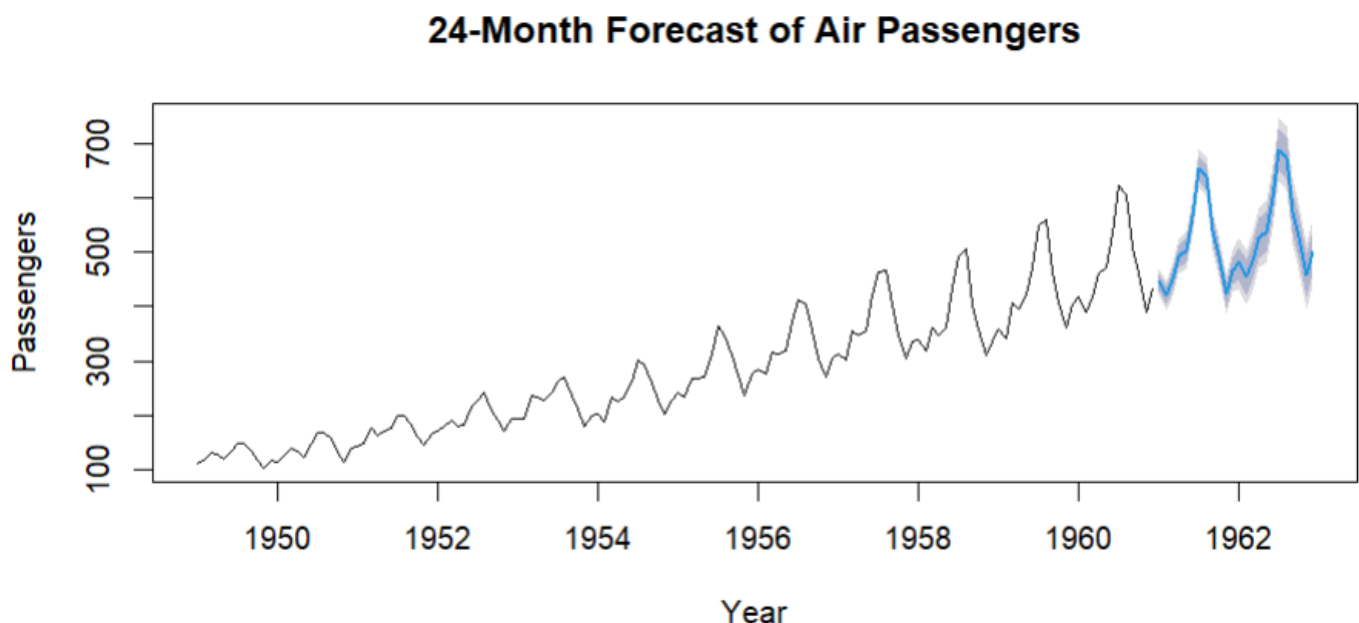
plot(stl_model)

model<-auto.arima(passengers_ts,
                  seasonal=TRUE)

forecast_result<-forecast(model,h=24)

plot(forecast_result,
     main="24-Month Passenger Forecast",
     xlab="Year",
     ylab="Passengers")
```

Interpretation



The STL decomposition shows a clear upward trend and strong yearly seasonality. Seasonal

fluctuations become larger as passenger numbers increase, so a multiplicative decomposition is more suitable.

The 24-month forecast continues the increasing trend and repeating seasonal pattern.

Result

STL successfully decomposed the series, and `auto.arima()` produced a 24-month forecast with prediction intervals.

QUESTION 3: Choropleth Map & Interactive Geospatial Plot

Part A — Choropleth Map

Question

Language: R

Dataset provided: USArrests.csv

Join USArrests.csv to US state boundary polygons and create a choropleth map of Murder rate by state using a sequential color scale. Add a legend, title and caption. Identify the state(s) with the highest murder rate and describe any regional clustering.

Aim

To visualize the geographical distribution of murder rates across US states.

R Code

```
library(ggplot2)
library(dplyr)
library(maps)

data<-read.csv("USArrests.csv",
               row.names=1)

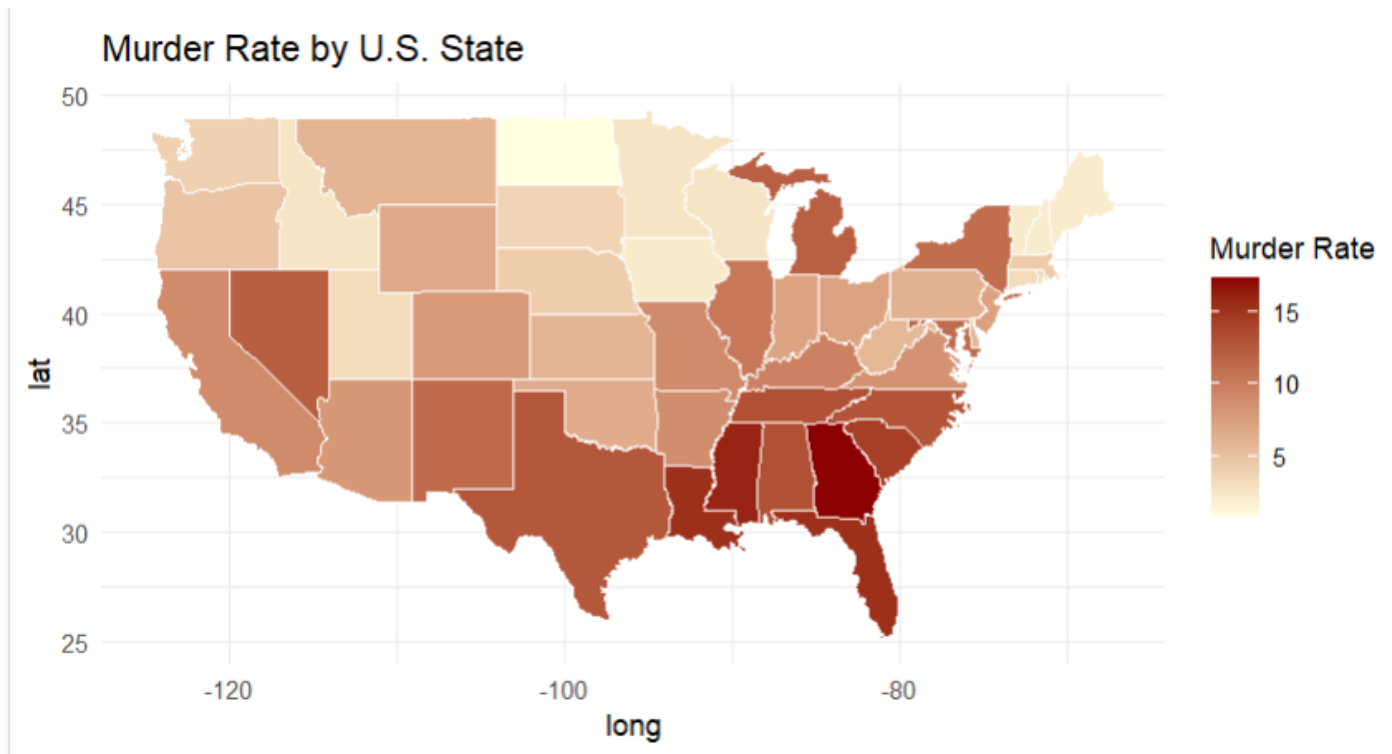
data$state<-tolower(rownames(data))

states<-map_data("state")

map_data_final<-states%>%
  left_join(data,by=c("region"="state"))

ggplot(map_data_final,
       aes(long,lat,group=group,fill=Murder))+
  geom_polygon(color="white")+
  coord_fixed(1.3)+
  scale_fill_gradient(low="lightyellow",
                     high="darkred")+
  labs(title="Murder Rate by U.S. State",
       fill="Murder Rate",
       caption="Source: USArrests")+
  theme_minimal()
```

Interpretation



The map shows considerable variation in murder rates between states. Several relatively high values are concentrated in the Southern region.

Result

A choropleth map was successfully created to show state-wise murder rates.

Part B — Interactive Geospatial Plot

Question

Language: R

Dataset provided: quakes.csv

Using the leaflet package, plot earthquake locations as circle markers, sizing each marker by magnitude and coloring by depth. Add popups showing magnitude and depth. Briefly describe any spatial clustering of high-magnitude events.

Aim

To visualize earthquake locations, magnitude and depth using an interactive map.

R Code

```
library(leaflet)
```

```
quakes<-read.csv("quakes.csv")
```

```
depth_palette<-colorNumeric(
  palette="YlOrRd",
  domain=quakes$depth
)
```

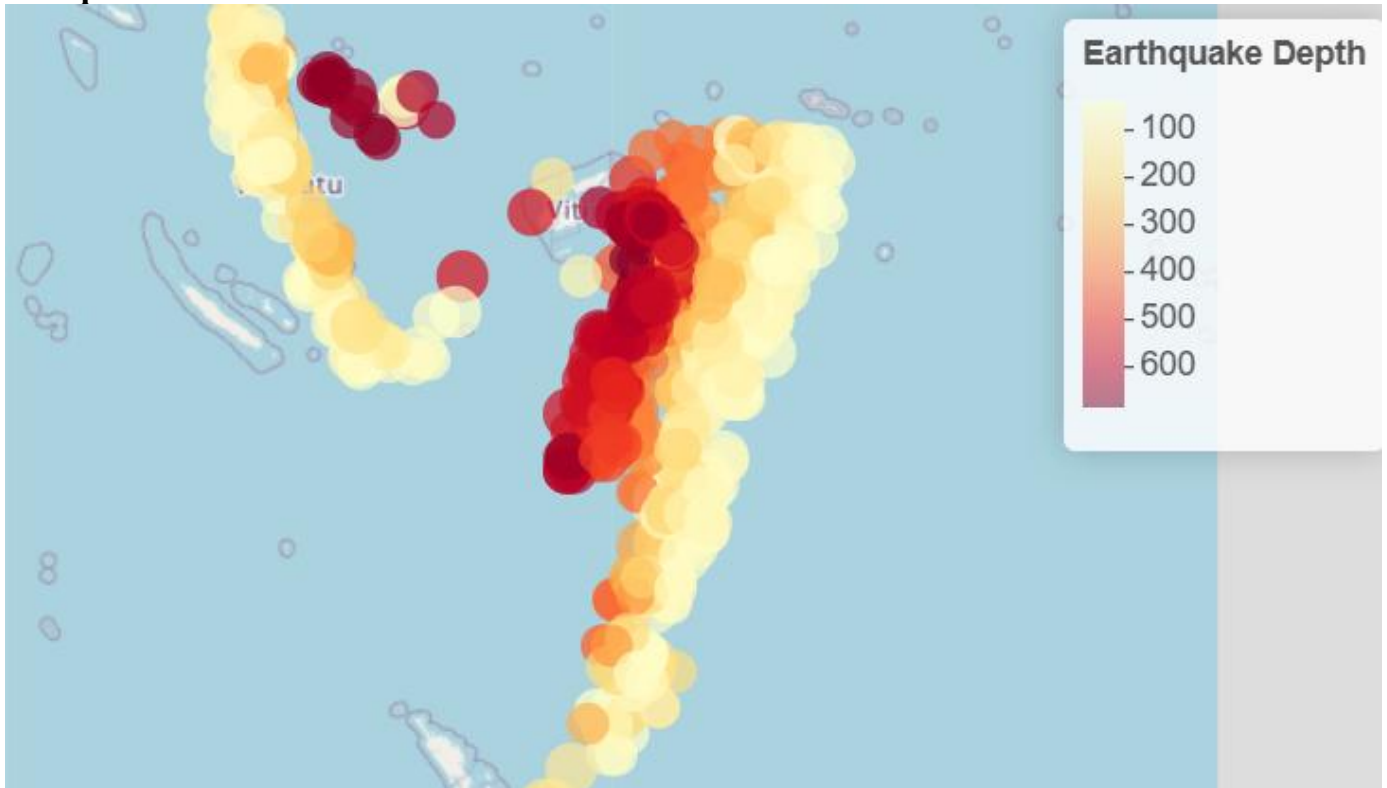
```
leaflet(quakes)%>%
  addTiles()%>%
  addCircleMarkers(
```

```

lng=~long,
lat=~lat,
radius=~mag*2,
color=~depth_palette(depth),
stroke=FALSE,
fillOpacity=0.7,
popup=~paste(
  "<b>Magnitude:</b>",mag,
  "<br><b>Depth:</b>",depth
)
)%>%
addLegend(
  pal=depth_palette,
  values=~depth,
  title="Depth"
)

```

Interpretation



The earthquake points show spatial clustering in the study region. Larger circles identify higher-magnitude earthquakes, while color represents depth.

Result

An interactive Leaflet earthquake map was successfully created with marker sizing, depth coloring and popups.

QUESTION 4: Cartogram with Map Projections

Question

Language: R

Dataset provided: state_x77.csv

Join state_x77.csv to US state boundary polygons and construct a population-weighted cartogram using the cartogram package. Render the cartogram alongside a standard choropleth of Population, using two different projections. Discuss how the cartogram changes the visual impression of state size and how projection affects the standard map's shape.

Aim

To compare a normal population map with a population-weighted cartogram and understand the effect of map projections.

R Code

```
library(sf)
library(ggplot2)
library(dplyr)
library(cartogram)
library(USAboundaries)

data<-read.csv("state_x77.csv")

data$state<-tolower(rownames(data))

us_states<-us_states(map_date="latest")

us_states$state<-tolower(us_states$name)

states_joined<-us_states%>%
  left_join(data,by="state")

population_cartogram<-cartogram_cont(
  states_joined,
  weight="Population",
  itermax=10
)

ggplot(states_joined)+
  geom_sf(aes(fill=Population))+
  scale_fill_viridis_c()+
  coord_sf(crs=4326)+
  labs(title="Population Map - Longitude/Latitude")+
  theme_minimal()

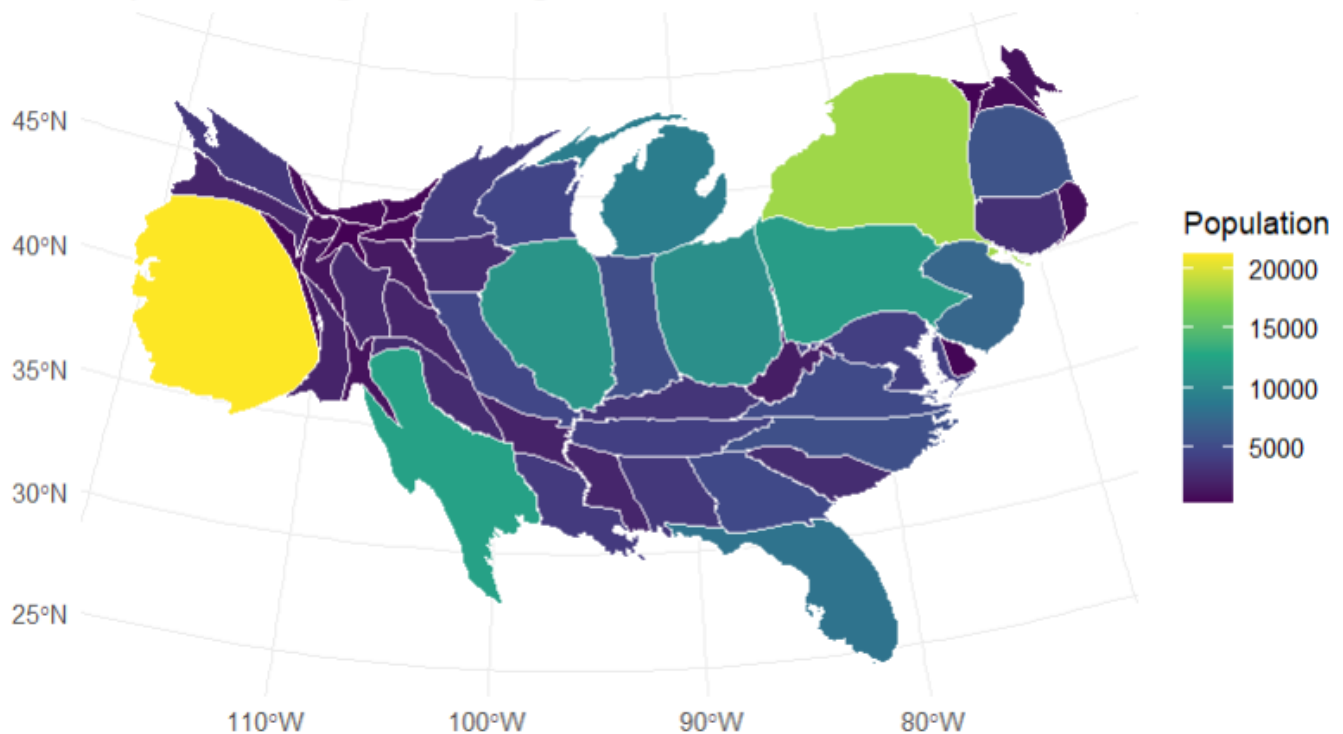
ggplot(states_joined)+
  geom_sf(aes(fill=Population))+
  scale_fill_viridis_c()+
  coord_sf(crs=5070)+
  labs(title="Population Map - Albers Projection")+
  theme_minimal()
```



```
ggplot(population_cartogram)+
  geom_sf(aes(fill=Population))+
  scale_fill_viridis_c()+
  labs(title="Population-Weighted Cartogram")+
  theme_minimal()
```

Interpretation

Population-Weighted Cartogram



In a normal map, states retain their actual geographical size. In a cartogram, states with larger populations become visually larger, while less populated states become smaller.

The Albers equal-area projection provides a different shape representation from a simple longitude/latitude projection and is more suitable when comparing geographic areas.

Result

The cartogram successfully represents population concentration through distorted state areas, while the standard maps preserve geographic representation.

QUESTION 5: 3D Geospatial Heatmap

Question

Language: Python

Dataset provided: volcano_elevation.csv

Load volcano_elevation.csv, pivot the data into a 2D elevation grid and create an interactive 3D surface heatmap using Plotly. Add axis titles and a title, and enable hover tooltips showing elevation at each grid point. Additionally produce a 2D top-down heatmap of the same data for comparison. Discuss what the 3D surface view reveals about the volcano's shape that the 2D heatmap alone does not.

Aim

To visualize the elevation and terrain structure of the volcano using 3D and 2D heatmaps.

Python Code

```

import pandas as pd
import plotly.graph_objects as go
import matplotlib.pyplot as plt

df=pd.read_csv("volcano_elevation.csv")

grid=df.pivot(index="y",
               columns="x",
               values="elevation")

fig=go.Figure(
    data=[
        go.Surface(
            z=grid.values,
            x=grid.columns,
            y=grid.index,
            colorscale="Earth",
            hovertemplate=
            "X: %{x}<br>"+
            "Y: %{y}<br>"+
            "Elevation: %{z}<extra></extra>"
        )
    ]
)

fig.update_layout(
    title="3D Surface Heatmap of Maunga Whau Volcano",
    scene=dict(
        xaxis_title="X",
        yaxis_title="Y",
        zaxis_title="Elevation"
    )
)

fig.show()

plt.figure(figsize=(10,7))

plt.imshow(
    grid.values,
    origin="lower",
    aspect="auto"
)

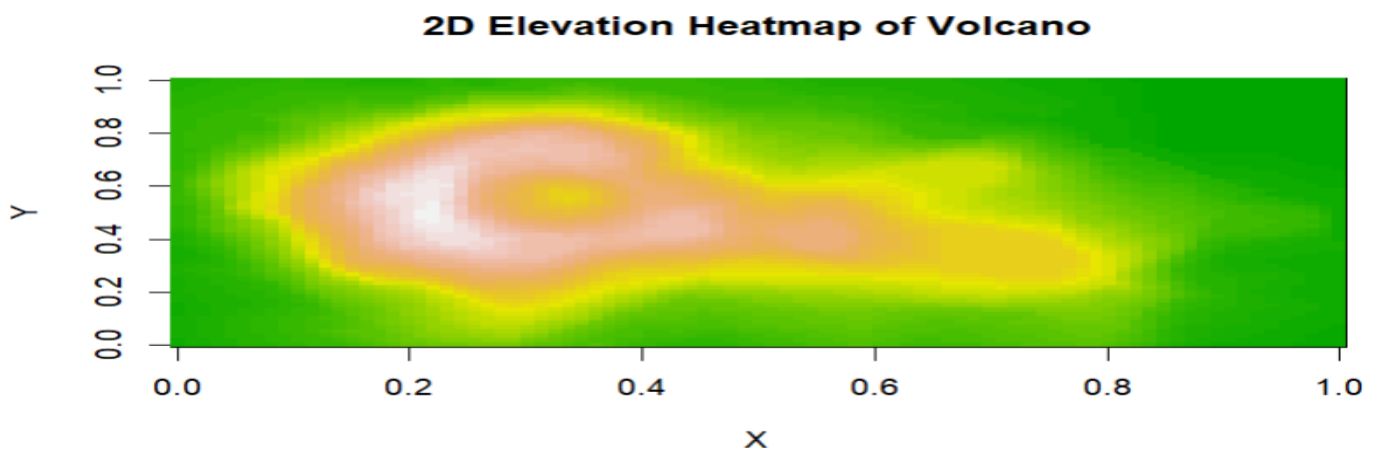
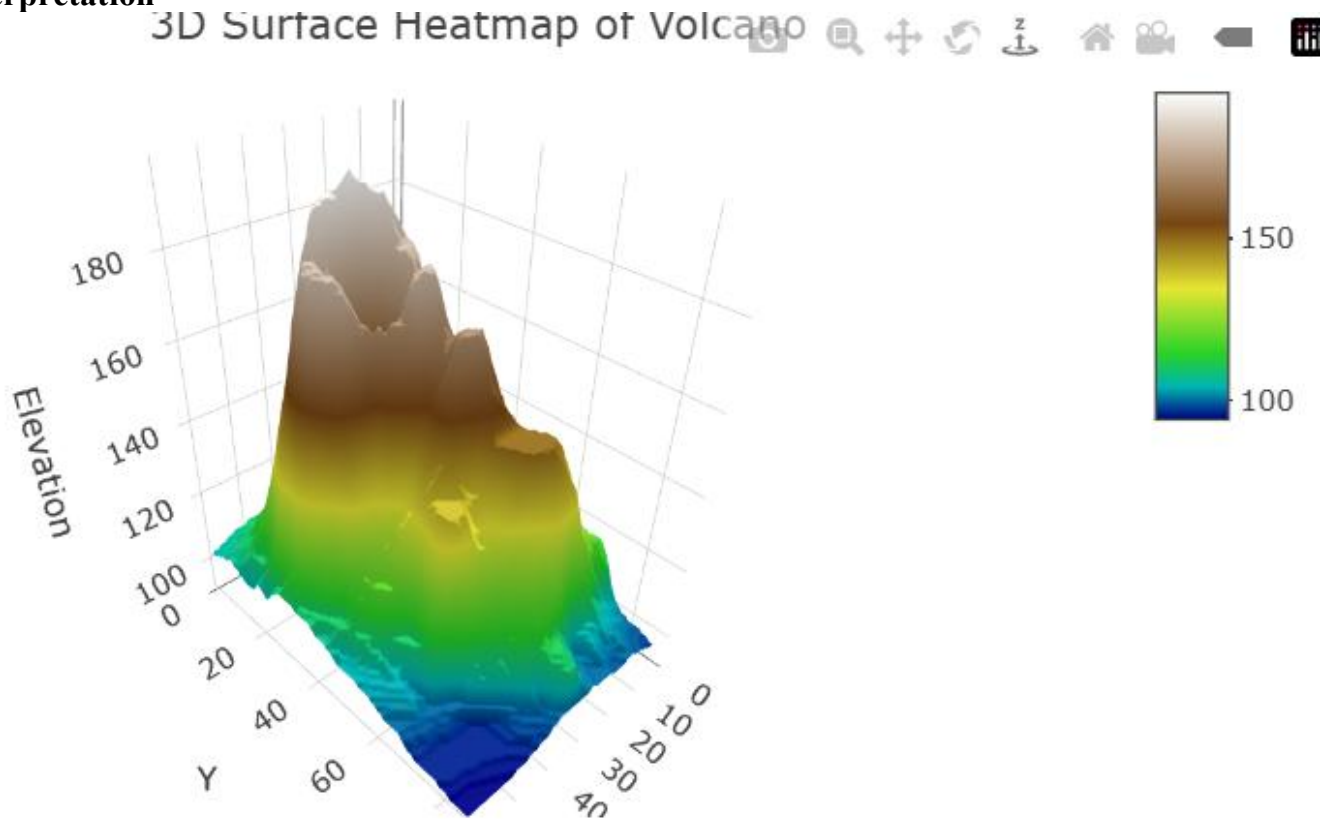
plt.colorbar(label="Elevation")
plt.title("2D Elevation Heatmap")

```

```
plt.xlabel("X")  
plt.ylabel("Y")
```

```
plt.show()
```

Interpretation



The 3D surface shows the height, peaks and slopes of the volcano more clearly. The 2D heatmap provides a top-down view and makes high- and low-elevation regions easy to compare, but it does not show the terrain shape as clearly.

Result

The elevation data were successfully converted into a 2D grid and visualized using both 3D Plotly surface visualization and a 2D heatmap.